

Robotics & AI Club

16-Class Curriculum Plan

16 classes per batch · 2 classes a week · 8 weeks

4 age pathways · 64 unique activities · 10 domains · capstone showcase

INTERNAL · FOR ACADEMIC LEADERSHIP



Little Explorers

Ages 5-7

Curious hands, big ideas



Junior Makers

Ages 8-11

From circuits to coded robots



Young Innovators

Ages 12-14

Real code, real electronics, real problems



Tech Leaders

Ages 15-17

Engineering solutions the world needs



CURRICULUM DEVELOPED BY

Qasim Mushtaq

MS Robotics and AI

Pak-Turk Maarif International Schools & Colleges, Islamabad Boys Campus, Chak Shahzad
Academic Year 2026-27 · Prepared September 2026



4

age pathways



16

classes per batch (2 a week, 8 weeks)



+2

optional capstone sessions after the batch



2

students per kit group

How every class runs (90-120 min)



1. Hook

A real-world question, video or demo (10 min)



2. Guided build

Step-by-step with the teacher; simulate first where possible



3. Discovery

Open challenge: change, extend or improve



4. Share

Show, explain and reflect (5-10 min)

Pathways and progression

Pathway	Classes 1-4 (Weeks 1-2)	Classes 5-8 (Weeks 3-4)	Classes 9-12 (Weeks 5-6)	Classes 13-16 (Weeks 7-8)	Mix (H/S/HS/U)
Little Explorers Ages 5-7	Circuits & Power Paper circuits, a doorbell, lemon batteries and a first traffic light program	Sensing the World Light, temperature, soil and motion sensors on pre-wired Arduino boards	Move & Communicate Waving robots, a railway gate, brush robots and Morse code messages	Code, Play & AI Makey Makey music, ScratchJr stories, a game controller and a first look at AI	12 / 1 / 2 / 1
Junior Makers Ages 8-11	Circuits, Code & Sensing Breadboards, multimeters, a reaction game, a weather station and a smart bin	Smart Home & Energy Rain, fire and gas safety, a smart fan and a sun-tracking solar panel	Security, Design & Robots A coded burglar alarm, 3D design, a line follower and a Bluetooth lamp	Health, Web, Games & AI A step counter, a first webpage, a knob-controlled game and an AI sorter	10 / 3 / 3 / 0
Young Innovators Ages 12-14	Electronics & Mechanisms 555 timers, heart-rate monitoring, energy saving and a drawing plotter	Robots & Security Scanning robots, a Bluetooth car, a gesture arm and an RFID lock	Power & Wireless Battery packs, radio doorbells, a Wi-Fi plant system and its phone app	Software, AI & Design Python games, a portfolio website, a chatbot and a 3D-printed enclosure	10 / 5 / 1 / 0
Tech Leaders Ages 15-17	Power, Autonomy & Health Off-grid solar IoT, a PID maze robot, a record-and-replay arm and a fall detector	Connected Systems ESP-NOW networks, a cloud smart home, smart parking and a two-factor locker	Data, AI & Web Data pipelines, machine learning, AI ethics and REST APIs	Security, Robots & Product Design IoT security, a fire-fighting robot, parametric CAD and a motion-controlled game	9 / 3 / 4 / 0

Design principles

- Every pathway covers all 10 domains, from electronics to AI; no activity is repeated across age groups.
- Simulate first (Tinkercad Circuits, Wokwi, PhET, Falstad), then build on real hardware.
- Blocks (5-11) to text code (12-17): mBlock/ArduBlock and ArduinoBlocks, then Arduino IDE, Python and web.
- Ages 5-7 use pre-wired Arduino boards and Makey Makey; ages 8+ wire their own circuits.

Safety and policy notes

- Ages 5-7: the teacher pre-wires every circuit; children plug, press and program with blocks.
- Flame and gas tests are teacher-only; relays switch low-voltage loads only.
- 18650 lithium cells (YI-09, TL-01, TL-04): protected cells, supervised charging only.
- Gas sensors need 24-48 h burn-in before the air quality classes.

ISTE Standards for Students
International Society for Technology in Education
Empowered learners, computational thinkers, innovative designers, digital citizens

CSTA PK-12 CS Standards (2026)
Computer Science Teachers Association
Algorithms, computing systems, data, networks and impacts, now with AI

NGSS Engineering Design
Next Generation Science Standards
Define a problem, design solutions, test and improve

UNESCO AI Competency Framework
UNESCO, 2024
Understand, apply and create with AI, with a human-centred and ethical mindset

Gold Standard PBL
PBLWorks
Real problems, student voice and choice, critique, revision and a public product

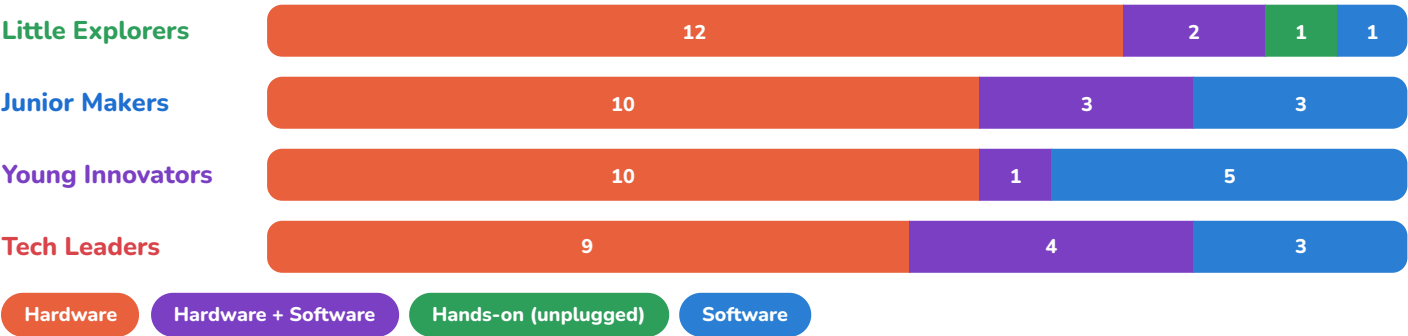
Plus UN SDGs
Every activity also carries UN Sustainable Development Goal badges (next page).

Domain coverage (number of classes touching each domain)

Domain	Little Explorers Ages 5-7	Junior Makers Ages 8-11	Young Innovators Ages 12-14	Tech Leaders Ages 15-17
Electronics & Circuits	2	1	2	1
Sensors & Environment	3	5	3	2
Robotics & Actuation	2	3	2	2
Mechanical Design	1	2	2	2
Power & Energy	1	1	2	1
Programming & Logic	3	1	2	2
IoT & Communication	1	1	3	4
Security & Safety	1	2	1	3
AI & Data	1	1	1	3
App, Web & Game Development	2	3	3	2

Counts include each activity's main and second domain. Every pathway covers all 10 domains.

Hands-on vs screen time



Each activity is tagged with the UN Sustainable Development Goals (SDGs) it helps students understand. Badges appear on every activity page, like achievements in a game, so students see how their builds connect to real global challenges.



SDG	Little Explorers Ages 5-7	Junior Makers Ages 8-11	Young Innovators Ages 12-14	Tech Leaders Ages 15-17	Example activities
SDG 2 · Zero Hunger	1	-	2	1	Plant Water Alarm; Wi-Fi Automatic Plant Watering System
SDG 3 · Good Health	1	4	2	2	Hot & Cold Temperature Indicator; Two-Player Reaction Timer with Scoreboard
SDG 4 · Quality Education	10	3	4	3	Light-Up Paper Circuit Card; Push-Button Doorbell
SDG 6 · Clean Water	-	-	1	-	Wi-Fi Automatic Plant Watering System
SDG 7 · Clean Energy	3	4	2	2	Light-Up Paper Circuit Card; Lemon Battery LED Lamp
SDG 8 · Decent Work	-	-	1	-	Personal Portfolio Website
SDG 9 · Industry & Innovation	6	3	10	10	Waving Robot Arm; Automatic Railway Crossing Gate
SDG 10 · Reduced Inequalities	1	-	-	1	Push-Button Doorbell; Fall Detection Wearable for the Elderly
SDG 11 · Sustainable Cities	4	5	3	6	Traffic Light Controller; Automatic Night Lamp
SDG 12 · Responsible Consumption	-	2	2	1	Smart Dustbin with Fill-Level Indicator; AI Recycling Sorter
SDG 13 · Climate Action	1	6	2	3	Hot & Cold Temperature Indicator; Mini Weather Station with OLED Display
SDG 15 · Life on Land	1	-	-	-	Plant Water Alarm
SDG 16 · Peace & Justice	1	1	1	3	Motion Detector Room Alarm; Home Burglar Alarm with Secret Code
SDG 17 · Partnerships	1	-	-	-	Morse Code Light Messenger

Badges are an internal teaching aid. Numbers and goal names follow the United Nations 2030 Agenda; badge artwork is our own.



16-class roadmap

8 weeks, 2 classes a week: four phases of 4 classes. The capstone showcase runs after the batch.

Phase 1

Weeks 1-2

Circuits & Power

Paper circuits, a doorbell, lemon batteries and a first traffic light program

C1



Light-Up Paper Circuit Card

Electronics & Circuits ·

Hardware

Electricity flows only in a closed loop

LE-01

C2



Push-Button Doorbell

Electronics & Circuits ·

Hardware

Pressing a button closes a circuit

LE-02

C3



Lemon Battery LED Lamp

Power & Energy · **Hardware**

Chemical energy can become electrical energy

LE-03

C4



Traffic Light Controller

Programming & Logic ·

Hardware

Sequences: instructions run in order

LE-04

Phase 2

Weeks 3-4

Sensing the World

Light, temperature, soil and motion sensors on pre-wired Arduino boards

C5



Automatic Night Lamp

Sensors & Environment ·

Hardware

Light and darkness can be measured

LE-05

C6



Hot & Cold Temperature Indicator

Sensors & Environment ·

Hardware

Hot and cold can be measured

LE-06

C7



Plant Water Alarm

Sensors & Environment ·

Hardware

Plants need water to live

LE-07

C8



Motion Detector Room Alarm

Security & Safety · **Hardware**

Our bodies give off heat

LE-08

Phase 3

Weeks 5-6

Move & Communicate

Waving robots, a railway gate, brush robots and Morse code messages

C9



Waving Robot Arm

Robotics & Actuation ·

Hardware

Motors create movement

LE-09

C10



Automatic Railway Crossing Gate

Robotics & Actuation ·

Hardware

Sensors, decisions and motors working together

LE-10

C11



Vibrating Brush Robot

Mechanical Design ·

Hardware

Balance and weight change movement

LE-11

C12



Morse Code Light Messenger

IoT & Communication ·

Hardware

Information can be sent as patterns (codes)

LE-12

Phase 4

Weeks 7-8

Code, Play & AI

Makey Makey music, ScratchJr stories, a game controller and a first look at AI

C13



Makey Makey Fruit Piano

Programming & Logic ·

Hardware + Software

Conductors and insulators

LE-13

C14



Animated Story App

App, Web & Game

Development · **Software**

Sequences and debugging

LE-14

C15



Catch-the-Fruit Game with Playdough Controller

App, Web & Game

Development · **Hardware + Software**

Events and scores in a game

LE-15

C16



Robot Sorting Rules Game

AI & Data · **Hands-on (unplugged)**

Machines sort data using rules

LE-16



After class 16 • Optional capstone showcase (outside the 16-class batch)

Suggested projects: Light-up invention for my bedroom; Smart plant pot helper; Makey Makey musical instrument; Talking robot face with lights and sounds. Teams choose a problem, plan, build, get feedback, improve and present at a showcase.

CLASS

1

Light-Up Paper Circuit Card
 Electronics & Circuits • Hardware • Week 1

LE-01

Introduction	Every electrical device, from a torch to a phone, works because electricity travels around a complete loop called a circuit. In this first session children draw a path with copper tape on a greeting card and place an LED and a coin cell on it. When the loop is closed the light switches on; when it is broken the light goes off. They also discover that the LED only works one way round, their first taste of polarity. The finished card goes home as a gift.	 Software No software needed Also: PhET Circuit Construction Kit (teacher demo) Key hardware (per group) - LED 5 mm (assorted) x2 - Coin cell CR2032 x1 Standards NGSS K-2-ETS1 • ISTE Empowered Learner SDG badges 4 Quality Education 7 Clean Energy
Activities	GUIDED - Draw the circuit path and stick copper tape along it - Place the LED and coin cell the correct way round DISCOVERY - Add a paper-flap switch so the light turns on when the card opens	
Outcomes	Science: Electricity flows only in a closed loop Technology: Batteries have a + and a - side Arts: Designing a personalised card	

CLASS

2

Push-Button Doorbell
 Electronics & Circuits • Hardware • Week 1

LE-02

Introduction	When a visitor presses a doorbell, the button closes a circuit and a sound plays inside the house. Children explore this everyday invention by pressing a button on a pre-wired Arduino board to sound a buzzer. They then use picture blocks to choose how long the bell rings and add a flashing light for people who cannot hear the sound. The session links a button (input) to a sound (output), the pattern behind every machine they will build.	 Software mBlock 5 / ArduBlock (picture blocks) Also: Tinkercad Circuits (teacher demo) Key hardware (per group) - Arduino Uno + USB cable x1 - Push button x1 - Buzzer x1 - LED 5 mm (assorted) x1 - Resistor 220 ohm x1 - Resistor 10k ohm x1 Standards NGSS K-2-ETS1 • CSTA Computing Systems SDG badges 4 Quality Education 10 Reduced Inequalities
Activities	GUIDED - Press the button to sound the buzzer - Use picture blocks: when button pressed, play a beep DISCOVERY - Add a flashing LED so deaf visitors also know someone is at the door	
Outcomes	Science: Pressing a button closes a circuit Technology: Inputs and outputs Life Skills: Designing for people with different needs	

CLASS
3

Lemon Battery LED Lamp

Power & Energy • Hardware • Week 2

LE-03

Introduction

Batteries store chemical energy and turn it into electricity. In this hands-on science lab children push zinc nails and copper coins into lemons to make their own battery. One lemon is too weak, so they connect several lemons in a chain and watch an LED begin to glow. They compare results, count lemons and talk about where the energy really comes from.



Software

No software needed

Key hardware (per group)

- LED 5 mm (assorted) x1

Standards

NGSS K-2-ETS1 • ISTE Knowledge Constructor

SDG badges



Activities

GUIDED

- Insert a zinc nail and a copper coin into each lemon
- Connect three or four lemons in a chain with clips to light an LED

DISCOVERY

- Which works better: 2 lemons or 4 lemons? Test and share

Outcomes

Science: Chemical energy can become electrical energy

Maths: Counting and comparing results

Engineering: Connecting cells in series

CLASS
4

Traffic Light Controller

Programming & Logic • Hardware • Week 2

LE-04

Introduction

Traffic lights keep cars and people safe by switching colours in a fixed order. Children snap together picture blocks that switch a red, yellow and green LED on and off in the right sequence on a real Arduino. They discover that a computer follows instructions exactly, in order, and that changing one wait block changes the whole pattern. They finish by designing a safe crossing time for children at the school gate.



Software

mBlock 5 / ArduBlock (picture blocks)

Also: Tinkercad Circuits (teacher demo)

Key hardware (per group)

- Arduino Uno + USB cable x1
- LED 5 mm (assorted) x3
- Resistor 220 ohm x3
- Buzzer x1

Standards

CSTA Algorithms & Programming • ISTE Computational Thinker

SDG badges



Activities

GUIDED

- Put the light blocks in the correct order
- Use wait blocks to set how long each colour stays on

DISCOVERY

- Add a buzzer that beeps while the green man is walking

Outcomes

Computing: Sequences: instructions run in order

Maths: Time in seconds

Life Skills: Road safety

CLASS

5

Automatic Night Lamp

Sensors & Environment • Hardware • Week 3

LE-05

Introduction	Streetlights across a city switch on by themselves when it gets dark, saving electricity and keeping roads safe. Children meet their first sensor, a light sensor, and see how the Arduino reads whether it is light or dark. Using an if-then picture block, they make an LED switch on when the sensor is covered. They test it with a torch and their hands and explain how the lamp knows when to turn on.	 Software mBlock 5 / ArduBlock (picture blocks) Key hardware (per group) • Arduino Uno + USB cable x1 • LDR module x1 • LED 5 mm (assorted) x2 • Resistor 220 ohm x2 Standards CSTA Algorithms & Programming • NGSS K-2-ETS1 SDG badges 7 Clean Energy 11 Sustainable Cities
Activities	GUIDED • Cover and uncover the light sensor and watch the reading • Use an if-then block to switch the LED on in the dark DISCOVERY • Make the lamp change colour at sunset	
Outcomes	Science: Light and darkness can be measured Computing: If-then decisions Life Skills: Saving energy	

CLASS

6

Hot & Cold Temperature Indicator

Sensors & Environment • Hardware • Week 3

LE-06

Introduction	Thermometers help us decide what to wear, when a child has a fever and when food is safe to eat. Children use a temperature sensor that lights a red LED when it is warm and a blue LED when it is cool. They warm the sensor in their hands, cool it near a cold bottle and watch the light change. They then choose the temperature that should count as hot, their first experience of setting a threshold.	 Software mBlock 5 / ArduBlock (picture blocks) Key hardware (per group) • Arduino Uno + USB cable x1 • DHT22 sensor x1 • LED 5 mm (assorted) x2 • Resistor 220 ohm x2 • Resistor 10k ohm x1 Standards NGSS K-2 Science Practices • CSTA Data & Analysis SDG badges 3 Good Health 13 Climate Action
Activities	GUIDED • Warm the sensor with hands and watch the light change • Choose the temperature where red turns on DISCOVERY • Test different places in the classroom: which is warmest?	
Outcomes	Science: Hot and cold can be measured Maths: Comparing bigger and smaller numbers Computing: Thresholds in decisions	

CLASS
7

Plant Water Alarm
Sensors & Environment • Hardware • Week 4

LE-07

Introduction	Plants cannot tell us when they are thirsty, but a sensor can. Children push a soil moisture sensor into a real plant pot and watch the reading change as they add water. They program a buzzer and LED to warn them when the soil is too dry, then care for the class plant through the term. The activity connects technology with nature and the farmers who grow our food.	 Software mBlock 5 / ArduBlock (picture blocks) Key hardware (per group) • Arduino Uno + USB cable x1 • Soil moisture sensor x1 • Buzzer x1 • LED 5 mm (assorted) x1 • Resistor 220 ohm x1 Standards NGSS K-2 LS1 • ISTE Empowered Learner SDG badges 2 Zero Hunger 15 Life on Land
Activities	GUIDED • Test the sensor in dry soil and in wet soil • Make the LED and buzzer warn when the soil is dry DISCOVERY • How much water makes the alarm stop?	
Outcomes	Science: Plants need water to live Technology: Sensors in farming Life Skills: Caring for living things	

CLASS
8

Motion Detector Room Alarm
Security & Safety • Hardware • Week 4

LE-08

Introduction	Security lights and alarms notice when someone moves nearby. Children discover the PIR motion sensor, which senses the warmth of moving bodies, and use it to sound a buzzer and flash a light. They play a 'sneak past the guard' game to test how well it works and find its blind spots. They finish by discussing how alarms keep homes and shops safe.	 Software mBlock 5 / ArduBlock (picture blocks) Key hardware (per group) • Arduino Uno + USB cable x1 • PIR motion sensor x1 • Buzzer x1 • LED 5 mm (assorted) x1 • Resistor 220 ohm x1 Standards NGSS K-2-ETS1 • ISTE Computational Thinker SDG badges 11 Sustainable Cities 16 Peace & Justice
Activities	GUIDED • Wave in front of the sensor and watch the LED respond • Program the buzzer to sound when motion is detected DISCOVERY • Can you sneak past the alarm? Where is its blind spot?	
Outcomes	Science: Our bodies give off heat Technology: Motion sensors in security Life Skills: Keeping homes safe	

CLASS
9

Waving Robot Arm

Robotics & Actuation • Hardware • Week 5

LE-09

Introduction	Robots in airports and hospitals greet people by waving and moving their arms. Children decorate a cardboard robot and attach its arm to a servo motor, a motor that turns to an exact position. With picture blocks they choose the angles, then press a button to make their robot wave hello. They learn that robots move because code tells motors exactly where to go.	Software mBlock 5 / ArduBlock (picture blocks) Key hardware (per group) • Arduino Uno + USB cable x1 • Servo SG90 x1 • Push button x1 • Resistor 10k ohm x1 Standards NGSS K-2-ETS1 • ISTE Innovative Designer SDG badges 9 Industry & Innovation 4 Quality Education
Activities	GUIDED • Attach a craft arm to the servo • Set two angles and repeat them to make a wave DISCOVERY • Make your robot wave faster, slower or do a dance	
Outcomes	Engineering: Motors create movement Maths: Turning and angles Arts: Designing a robot character	

CLASS
10

Automatic Railway Crossing Gate

Robotics & Actuation • Hardware • Week 5

LE-10

Introduction	At a railway crossing, a barrier lowers and a bell rings when a train comes near. Children build a model crossing: an ultrasonic sensor 'sees' an approaching toy train, a servo lowers a gate and a buzzer rings. When the train has passed, the gate lifts again. This is their first project that joins a sensor, a decision and a motor into one working system.	Software mBlock 5 / ArduBlock (picture blocks) Key hardware (per group) • Arduino Uno + USB cable x1 • Ultrasonic sensor HC-SR04 x1 • Servo SG90 x1 • Buzzer x1 • LED 5 mm (assorted) x2 • Resistor 220 ohm x2 Standards NGSS K-2-ETS1 • CSTA Computing Systems SDG badges 9 Industry & Innovation 11 Sustainable Cities
Activities	GUIDED • Measure distance with the ultrasonic sensor • Lower the gate and ring the bell when the train is close DISCOVERY • Add flashing red lights that blink while the gate is down	
Outcomes	Engineering: Sensors, decisions and motors working together Science: Echoes measure distance Life Skills: Railway safety	

CLASS
11

Vibrating Brush Robot
 Mechanical Design • Hardware • Week 6

LE-11

Introduction	Not every robot has wheels. Some move by shaking! Children build a tiny robot from a brush head and a vibration motor and discover how its shape, weight and balance change the way it moves. They race their robots on a plate arena and redesign them to go straighter or spin in circles. This is engineering design at its simplest: build, test, improve.	 Software No software needed Key hardware (per group) • Brush Bot kit x1 • Coin cell CR2032 x1 Standards NGSS K-2-ETS1-2 • ISTE Innovative Designer SDG badges 9 Industry & Innovation 4 Quality Education
Activities	GUIDED • Stick the motor and coin cell onto the brush head • Race your robot on the arena DISCOVERY • Move the motor: can you make the robot turn in circles?	
Outcomes	Engineering: Balance and weight change movement Science: Vibration is a kind of motion Life Skills: Improving after a test	

CLASS
12

Morse Code Light Messenger
 IoT & Communication • Hardware • Week 6

LE-12

Introduction	Before phones, people sent messages across oceans using dots and dashes called Morse code. Children learn the code for a few letters, then send secret messages to a partner using a button that flashes a light and beeps a buzzer. The partner decodes the message on a picture chart. They discover that all communication, even Wi-Fi, is information sent as patterns.	 Software mBlock 5 / ArduBlock (picture blocks) Key hardware (per group) • Arduino Uno + USB cable x1 • Push button x1 • LED 5 mm (assorted) x1 • Buzzer x1 • Resistor 220 ohm x1 • Resistor 10k ohm x1 Standards CSTA Networks & the Internet • ISTE Creative Communicator SDG badges 9 Industry & Innovation 17 Partnerships
Activities	GUIDED • Learn the Morse code for SOS and your initials • Send a message with the button, light and buzzer DISCOVERY • Send a secret word to another group across the room	
Outcomes	Computing: Information can be sent as patterns (codes) Science: Light and sound carry messages Life Skills: Listening and taking turns	

CLASS
13

Makey Makey Fruit Piano

Programming & Logic • Hardware + Software • Week 7

LE-13

Introduction

Almost anything that conducts electricity can become a computer key. Children connect bananas, apples and foil to a Makey Makey board and turn them into a working piano. They test which objects work and which do not, and discover that water in the fruit carries a tiny current. They finish by playing a simple tune together as a class band.

Activities

GUIDED

- Clip fruit to the Makey Makey keys
- Play notes on the piano app

DISCOVERY

- Which classroom objects work as keys? Sort them into two groups

Outcomes

Science: Conductors and insulators
Technology: Inputs control computers
Arts: Making music together



Software

Makey Makey Apps (piano)

Also: Scratch 3

Key hardware (per group)

- Makey Makey kit x1

Standards

CSTA Computing Systems • ISTE Empowered Learner

SDG badges



Introduction

Apps and cartoons are built from small instructions that tell characters what to do. Children play an unplugged 'program the teacher' game, then use picture blocks in ScratchJr to make characters move, jump and speak. They build a three-scene story about a robot helping their community. They learn to plan, sequence and fix mistakes (debugging) in their own app.

Activities

GUIDED

- Unplugged game: give step-by-step instructions to a friend
- Make a character move and speak with ScratchJr blocks

DISCOVERY

- Create a 3-scene story about a helpful robot

Outcomes

Computing: Sequences and debugging
Arts: Storytelling and character design
Life Skills: Planning ideas in order



Software

ScratchJr

Also: Scratch 3

Key hardware (per group)

- Tablet (1 per group)

Standards

CSTA Algorithms & Programming • ISTE Creative Communicator

SDG badges


CLASS
14

Animated Story App

App, Web & Game Development • Software • Week 7

LE-14

CLASS
15**Catch-the-Fruit Game with Playdough Controller**

App, Web & Game Development + Programming & Logic · Hardware + Software · Week 8

LE-15

Introduction

Game designers create both the game and the controller players hold. Children remix a simple Scratch game where a basket catches falling fruit, then build their own controller from playdough and a Makey Makey. They change the speed and score rules and test their game on classmates. It combines coding, design and hands-on building in one playful project.

**Software****Scratch 3**

Also: Makey Makey Apps

Key hardware (per group)

- Makey Makey kit x1

Standards

CSTA Algorithms & Programming · ISTE Innovative Designer

SDG badges**Activities****GUIDED**

- Build left and right playdough buttons with Makey Makey
- Remix the Scratch game: speed, score and sounds

DISCOVERY

- Invite a friend to test your game and improve one rule

Outcomes**Computing:** Events and scores in a game**Engineering:** Designing a controller**Arts:** Game characters and soundsCLASS
16**Robot Sorting Rules Game**

AI & Data · Hands-on (unplugged) · Week 8

LE-16

Introduction

Computers and AI sort information by asking simple yes-or-no questions, like a game of twenty questions. Children sort a pile of animal and object picture cards by making their own rules, such as 'Does it fly?' and 'Does it live in water?'. They draw their rules as a question tree, then a classmate acts as the 'robot' and follows the rules exactly to sort new cards. When the robot makes a mistake, children fix their rules, learning that machines are only as smart as the rules and data we give them.

**Software**

No software needed (unplugged)

Key hardware (per group)

- Picture cards and stationery

Standards

UNESCO AI Competency (Understand) · CSTA Data & Analysis

SDG badges**Activities****GUIDED**

- Sort picture cards by asking yes/no questions
- Draw your rules as a question tree

DISCOVERY

- A friend plays the robot: does your tree sort new cards correctly?

Outcomes**AI:** Machines sort data using rules**Computing:** Decision trees and patterns**Life Skills:** Clear instructions and fixing mistakes



16-class roadmap

8 weeks, 2 classes a week: four phases of 4 classes. The capstone showcase runs after the batch.

Phase 1

Weeks 1-2 Circuits, Code & Sensing

Breadboards, multimeters, a reaction game, a weather station and a smart bin

C1



Series & Parallel Circuit Lab

Electronics & Circuits · **Hardware + Software**
Current, voltage and circuit paths

JM-01

C2



Two-Player Reaction Timer with Scoreboard

Programming & Logic · **Hardware**
Variables, random numbers and timing

JM-02

C3



Mini Weather Station with OLED Display

Sensors & Environment · **Hardware**
Weather: temperature, humidity and light

JM-03

C4



Smart Dustbin with Fill-Level Indicator

Robotics & Actuation · **Hardware**
Combining two sensors and a motor

JM-04

Phase 2

Weeks 3-4 Smart Home & Energy

Rain, fire and gas safety, a smart fan and a sun-tracking solar panel

C5



Rain-Sensing Automatic Window

Sensors & Environment · **Hardware**
Water conducts electricity

JM-05

C6



Fire & Gas Safety Alarm Panel

Security & Safety · **Hardware**
Fire, smoke and gases

JM-06

C7



Temperature-Controlled Smart Fan

Robotics & Actuation · **Hardware**
Motor drivers and speed control

JM-07

C8



Sun-Tracking Solar Panel

Power & Energy · **Hardware**
The sun as an energy source

JM-08

Phase 3

Weeks 5-6 Security, Design & Robots

A coded burglar alarm, 3D design, a line follower and a Bluetooth lamp

C9



Home Burglar Alarm with Secret Code

Security & Safety · **Hardware**
Sequences, passwords and states

JM-09

C10



3D-Designed Name Badge & Sensor Mount

Mechanical Design · **Software**
Designing parts to measured sizes

JM-10

C11



Line Following Robot

Robotics & Actuation · **Hardware**
Robot chassis, motors and sensors

JM-11

C12



Phone-Controlled RGB Mood Lamp

IoT & Communication · **Hardware + Software**
Wireless messages between devices

JM-12

Phase 4

Weeks 7-8 Health, Web, Games & AI

A step counter, a first webpage, a knob-controlled game and an AI sorter

C13



Step Counter Fitness Tracker

Sensors & Environment · **Hardware**
Motion and acceleration

JM-13

C14



Weather Report Webpage

App, Web & Game Development · **Software**
HTML structure and CSS styling

JM-14

C15



Pong Game with a Knob Controller

App, Web & Game Development · **Hardware + Software**
Mapping values and game loops

JM-15

C16



AI Recycling Sorter

AI & Data · **Software**
Training data and accuracy

JM-16



After class 16 • Optional capstone showcase (outside the 16-class batch)

Suggested projects: Automatic pet feeder with a timer; Smart study lamp with posture reminder; Obstacle avoiding delivery car; Classroom noise-level meter. Teams choose a problem, plan, build, get feedback, improve and present at a showcase.

CLASS
1**Series & Parallel Circuit Lab**

Electronics & Circuits • Hardware + Software • Week 1

JM-01

Introduction

Why do some fairy lights all go out when one bulb breaks, while house lights keep working? Students simulate series and parallel circuits in PhET, then build both on a breadboard with LEDs, switches and a battery. With a multimeter they measure voltage in each circuit and record what they see. This lab gives them the wiring and measuring skills they will use for the rest of the programme.

Activities

GUIDED

- Build both circuits in the PhET simulator
- Wire them on a breadboard and measure voltage with a multimeter

DISCOVERY

- Remove one LED from each circuit: which keeps working and why?

Outcomes

Science: Current, voltage and circuit paths**Technology:** Using a multimeter safely**Maths:** Recording and comparing measurements**Software****PhET Circuit Construction Kit**

Also: Tinkercad Circuits

Key hardware (per group)

- LED 5 mm (assorted) x4
- Resistor 220 ohm x4
- Slide switch x2
- 9V battery + snap x1

Standards

NGSS 3-5-ETS1 • CSTA Computing Systems

SDG badgesCLASS
2**Two-Player Reaction Timer with Scoreboard**

Programming & Logic • Hardware • Week 1

JM-02

Introduction

Athletes, pilots and gamers all rely on fast reactions. Students build a two-player game: after a random wait an LED lights up, and the first player to press their button wins. The Arduino measures each reaction in milliseconds and shows the score on an OLED screen, while a buzzer marks false starts. They learn about random numbers, timing and keeping score with variables.

Activities

GUIDED

- Wire two buttons, two LEDs, a buzzer and the OLED
- Use blocks to add a random wait and measure time in ms

DISCOVERY

- Add a best-of-five match with a winner screen

Outcomes

Computing: Variables, random numbers and timing**Maths:** Milliseconds and comparing times**Life Skills:** Fair play**Software****ArduinoBlocks**

Also: Wokwi; Arduino IDE 2

Key hardware (per group)

- Arduino Uno + USB cable x1
- Push button x2
- LED 5 mm (assorted) x2
- Resistor 220 ohm x2
- Resistor 10k ohm x2
- Buzzer x1
- OLED 0.96 in I2C x1

Standards

CSTA Algorithms & Programming • ISTE Computational Thinker

SDG badges

CLASS
3

Mini Weather Station with OLED Display

Sensors & Environment • Hardware • Week 2

JM-03

Introduction

Weather stations help farmers, pilots and families plan their day. Students combine a temperature and humidity sensor with a light sensor and display all three readings on an OLED screen with simple icons. They take readings indoors, outdoors and in the shade and record them in a table. They learn how one microcontroller can read several sensors and present data clearly.



Software

ArduinoBlocks

Also: Wokwi (DHT22 + OLED)

Key hardware (per group)

- Arduino Uno + USB cable x1
- DHT22 sensor x1
- LDR module x1
- OLED 0.96 in I2C x1
- Resistor 10k ohm x1

Standards

CSTA Data & Analysis • NGSS 3-ESS2

SDG badges



Activities

GUIDED

- Read the DHT22 and the LDR
- Show temperature, humidity and light on the OLED

DISCOVERY

- Record readings in three places and explain the differences

Outcomes

Science: Weather: temperature, humidity and light**Computing:** Reading multiple sensors**Maths:** Tables and comparing dataCLASS
4

Smart Dustbin with Fill-Level Indicator

Robotics & Actuation • Hardware • Week 2

JM-04

Introduction

Smart bins in modern cities open without touching and tell cleaners when they are full. Students build one: an ultrasonic sensor outside detects a hand and a servo opens the lid, while a second ultrasonic sensor inside measures how full the bin is and lights a three-LED level bar. They build the bin from a cardboard box and test it with real rubbish. It is their first system with two sensors working at the same time.



Software

ArduinoBlocks

Also: Tinkercad Circuits; Wokwi

Key hardware (per group)

- Arduino Uno + USB cable x1
- Ultrasonic sensor HC-SR04 x2
- Servo SG90 x1
- LED 5 mm (assorted) x3
- Resistor 220 ohm x3

Standards

NGSS 3-5-ETS1 • ISTE Innovative Designer

SDG badges



Activities

GUIDED

- Open the lid with a servo when a hand is detected
- Measure fill level with the second sensor and light the LED bar

DISCOVERY

- Add a buzzer that beeps when the bin is full

Outcomes

Engineering: Combining two sensors and a motor**Computing:** Conditions with ranges**Life Skills:** Hygiene and clean cities

CLASS
5

Rain-Sensing Automatic Window

Sensors & Environment + Mechanical Design • Hardware • Week 3

JM-05

Introduction

Nobody wants rain soaking the room when a window is left open. Students build a model window that closes itself: a rain sensor detects water drops, a servo shuts the window and an LED and buzzer alert the family. They test it with a spray bottle and then design a way to reopen the window when the rain stops. This links weather sensing with a real mechanism.



Software

ArduinoBlocks

Also: Tinkercad Circuits

Key hardware (per group)

- Arduino Uno + USB cable x1
- Rain sensor x1
- Servo SG90 x1
- Buzzer x1
- LED 5 mm (assorted) x1
- Resistor 220 ohm x1

Standards

NGSS 3-5-ETS1 • CSTA Computing Systems

SDG badges



Activities

GUIDED

- Test the rain sensor with drops of water
- Close the model window with a servo when rain is detected

DISCOVERY

- Reopen the window automatically when the sensor dries

Outcomes

Science: Water conducts electricity**Engineering:** Hinges and mechanisms**Computing:** Sensor-triggered actionsCLASS
6

Fire & Gas Safety Alarm Panel

Security & Safety + Sensors & Environment • Hardware • Week 3

JM-06

Introduction

Fire and gas leaks are dangerous at home and in factories, so buildings use detectors that warn people early. Students build a safety panel with a flame sensor and a gas sensor, a green 'SAFE' and red 'DANGER' light, a buzzer and an OLED status message. The teacher tests it safely with a candle and a gas lighter (unlit). Students also learn what to do in a real emergency.



Software

ArduinoBlocks

Also: Wokwi (MQ-2)

Key hardware (per group)

- Arduino Uno + USB cable x1
- Flame sensor x1
- MQ-2 gas sensor x1
- Buzzer x1
- LED 5 mm (assorted) x2
- Resistor 220 ohm x2
- OLED 0.96 in I2C x1

Standards

NGSS 3-5-ETS1 • ISTE Digital Citizen

SDG badges



Activities

GUIDED

- Read the flame sensor and the MQ-2 gas sensor
- Show SAFE or DANGER on the OLED with lights and a buzzer

DISCOVERY

- Design the alarm so a flashing light also helps people who cannot hear

Outcomes

Science: Fire, smoke and gases**Technology:** Safety sensors in buildings**Life Skills:** Emergency response

CLASS
7

Temperature-Controlled Smart Fan

Robotics & Actuation • Hardware • Week 4

JM-07

Introduction

Air conditioners and cooling fans decide how fast to run by reading the temperature. Students build a smart fan: a temperature sensor measures the room, a motor driver controls a DC motor's speed, and an OLED shows the temperature and speed. A knob lets them switch to manual mode. They learn why motors need a driver and how automatic control saves energy.



Software

ArduinoBlocks

Also: Arduino IDE 2

Key hardware (per group)

- Arduino Uno + USB cable x1
- DHT22 sensor x1
- DC hobby motor x1
- L298N motor driver x1
- Potentiometer 10k x1
- OLED 0.96 in I2C x1
- 4xAA battery pack x1

Standards

NGSS 3-5-ETS1 • CSTA Computing Systems

SDG badges



Activities

GUIDED

- Drive a DC motor through the L298N
- Make fan speed rise as the temperature rises

DISCOVERY

- Add a manual mode with the potentiometer

Outcomes

Engineering: Motor drivers and speed control**Science:** Heat and cooling**Computing:** Mapping one range to anotherCLASS
8

Sun-Tracking Solar Panel

Power & Energy + Sensors & Environment • Hardware • Week 4

JM-08

Introduction

Solar farms produce more electricity when their panels follow the sun across the sky. Students build a sun tracker: two light sensors compare light on each side and a servo turns the solar panel towards the brighter side. They measure the panel's output with and without tracking using a multimeter. This connects clean energy, sensing and motion in one project.



Software

ArduinoBlocks

Also: Tinkercad Circuits

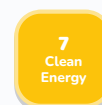
Key hardware (per group)

- Arduino Uno + USB cable x1
- LDR module x2
- Servo SG90 x1
- Small solar panel x1
- LED 5 mm (assorted) x1

Standards

NGSS 3-5-ETS1 • ISTE Innovative Designer

SDG badges



Activities

GUIDED

- Compare two LDR readings
- Turn the servo towards the brighter side

DISCOVERY

- Measure panel voltage with and without tracking

Outcomes

Science: The sun as an energy source**Engineering:** Feedback: sense, compare, move**Maths:** Comparing measurements

CLASS
9

Home Burglar Alarm with Secret Code

Security & Safety • Hardware • Week 5

JM-09

Introduction

Real alarm systems can be switched on and off with a secret code so the family is not woken by their own alarm. Students build a burglar alarm with a motion sensor and add three buttons that must be pressed in the correct order to disarm it. LEDs show whether the alarm is armed and a buzzer sounds if someone enters. They learn about passwords, order and security logic.



Software

ArduinoBlocks

Also: Arduino IDE 2

Key hardware (per group)

- Arduino Uno + USB cable x1
- PIR motion sensor x1
- Push button x3
- Resistor 10k ohm x3
- Buzzer x1
- LED 5 mm (assorted) x2
- Resistor 220 ohm x2

Standards

CSTA Algorithms & Programming • ISTE Digital Citizen

SDG badges



Activities

GUIDED

- Arm the alarm and detect motion with the PIR
- Program a 3-button secret code to disarm it

DISCOVERY

- Add a lockout after three wrong codes

Outcomes

Computing: Sequences, passwords and states**Technology:** Home security systems**Life Skills:** Keeping codes privateCLASS
10

3D-Designed Name Badge & Sensor Mount

Mechanical Design • Software • Week 5

JM-10

Introduction

Engineers design parts on computers before making them. Students learn Tinkercad 3D by making a name badge, then design a holder for an ultrasonic sensor that will be mounted on their robot. They measure the real sensor with a ruler, add holes of the right size and prepare the file for printing. The best designs are 3D printed and used in the next robot build.



Software

Tinkercad 3D

Also: Tinkercad Codeblocks; UltiMaker Cura

Key hardware (per group)

- Laptop (1 per group)

Standards

NGSS 3-5-ETS1 • ISTE Innovative Designer

SDG badges



Activities

GUIDED

- Combine shapes and holes to make a name badge
- Measure the sensor and design a mount that fits

DISCOVERY

- Improve your mount after test-fitting the printed part

Outcomes

Engineering: Designing parts to measured sizes**Maths:** Millimetres and 3D shapes**Arts:** Product design

CLASS
11

Line Following Robot

Robotics & Actuation • Hardware • Week 6

JM-11

Introduction

Warehouse and hospital robots follow lines on the floor to deliver goods and medicine. Students assemble a two-wheel robot with two IR sensors and a motor driver, then program it to steer left or right to stay on a black line. They tune speeds and test on a track, discovering how small changes affect the robot. This is their first complete autonomous robot.



Software

ArduinoBlocks

Also: mBlock 5; Arduino IDE 2

Key hardware (per group)

- Arduino Uno + USB cable x1
- 2WD robot chassis kit x1
- IR sensor module x2
- L298N motor driver x1
- 4xAA battery pack x1

Standards

CSTA Algorithms & Programming • NGSS 3-5-ETS1

SDG badges



Activities

GUIDED

- Assemble the chassis, motors, driver and IR sensors
- Program left, right and forward decisions

DISCOVERY

- Race the track: fastest robot that stays on the line

Outcomes

Engineering: Robot chassis, motors and sensors**Computing:** Decisions inside a loop**Technology:** Robots in industryCLASS
12

Phone-Controlled RGB Mood Lamp

IoT & Communication + App, Web & Game Development • Hardware + Software • Week 6

JM-12

Introduction

Smart bulbs at home can change colour from a phone. Students build a mood lamp with an RGB LED and a Bluetooth module, then use a ready-made MIT App Inventor app to send colour commands from an Android phone. They customise the app with their own buttons and a buzzer 'goodnight' tune. They discover how devices talk wirelessly by sending small messages.



Software

ArduinoBlocks + MIT App Inventor

Also: Serial Bluetooth Terminal

Key hardware (per group)

- Arduino Uno + USB cable x1
- Bluetooth module HC-05 x1
- RGB LED x1
- Resistor 220 ohm x3
- Resistor 1k ohm x1
- Resistor 10k ohm x1
- Buzzer x1

Standards

CSTA Networks & the Internet • ISTE Innovative Designer

SDG badges



Activities

GUIDED

- Mix colours on the RGB LED
- Pair with the phone app and send colour commands

DISCOVERY

- Add your own colour button and a bedtime mode in the app

Outcomes

Computing: Wireless messages between devices**Science:** Colour mixing with light**Technology:** Smart homes

CLASS
13

Step Counter Fitness Tracker

Sensors & Environment • Hardware • Week 7

JM-13

Introduction

Smartwatches count our steps using a tiny motion sensor. Students connect an accelerometer to an Arduino, see how its numbers change as they move, and write a rule that counts a step each time the movement passes a threshold. The step total shows on an OLED and resets with a button. They test their tracker on a walk around the school and compare with a friend's count.



Software

ArduinoBlocks

Also: Wokwi (MPU6050 + OLED)

Key hardware (per group)

- Arduino Uno + USB cable x1
- MPU6050 IMU x1
- OLED 0.96 in I2C x1
- Push button x1
- Resistor 10k ohm x1

Standards

CSTA Data & Analysis • NGSS 3-5-ETS1

SDG badges



Activities

GUIDED

- Watch accelerometer values change as you move
- Count steps with a threshold and show them on the OLED

DISCOVERY

- Improve accuracy: compare your count with 50 real steps

Outcomes

Science: Motion and acceleration**Computing:** Counting with thresholds**Life Skills:** Healthy activityCLASS
14

Weather Report Webpage

App, Web & Game Development • Software • Week 7

JM-14

Introduction

Every website is written in languages called HTML and CSS. Students build their first webpage: a weather report showing the readings from their JM-03 weather station, with headings, images and colours. They learn how tags structure a page and how CSS makes it look good. The finished page is shared with the class on the projector.

CLASS
15

Pong Game with a Knob Controller

App, Web & Game Development • Hardware + Software • Week 8

JM-15

Introduction

Pong was one of the first video games, controlled by a simple knob. Students recreate it in Scratch and build a real controller from a potentiometer and a button connected to an Arduino through mBlock. Turning the knob moves the paddle and the button serves the ball. They learn how physical inputs become numbers inside a game.



Software

mBlock 5 (Arduino + sprites)

Also: Scratch 3

Key hardware (per group)

- Arduino Uno + USB cable x1
- Potentiometer 10k x1
- Push button x1
- Resistor 10k ohm x1

Standards

CSTA Algorithms & Programming • ISTE Innovative Designer

SDG badges



Activities

GUIDED

- Wire the potentiometer and button
- Map the knob value to the paddle position in the game

DISCOVERY

- Add a two-player mode or faster ball after each point

Outcomes

Computing: Mapping values and game loops**Engineering:** Building a controller**Arts:** Game designCLASS
16

AI Recycling Sorter

AI & Data • Software • Week 8

JM-16

Introduction

Recycling plants are starting to use AI cameras to sort plastic, paper and metal. Students train an image AI in Teachable Machine using real recyclable items and a webcam, then test how accurately it sorts new items. They discover that poor or unbalanced examples make a biased AI, and fix it by collecting better data. They link AI to protecting the environment.



Software

Teachable Machine

Key hardware (per group)

- Laptop (1 per group)

Standards

UNESCO AI Competency (Understand/Apply) • CSTA AI concepts

SDG badges



Activities

GUIDED

- Collect example photos of plastic, paper and metal
- Train and test the model, and note its mistakes

DISCOVERY

- Improve accuracy by adding better, more varied examples

Outcomes

AI: Training data and accuracy**Science:** Materials and recycling**Life Skills:** Fairness: biased data makes biased AI



16-class roadmap

8 weeks, 2 classes a week: four phases of 4 classes. The capstone showcase runs after the batch.

Phase 1

Weeks 1-2

Electronics & Mechanisms

555 timers, heart-rate monitoring, energy saving and a drawing plotter

C1



555 Timer Alarm Siren Circuit

Electronics & Circuits · **Hardware**

Capacitors, resistance and charging time

YI-01

C2



Heart Rate Monitoring System

Sensors & Environment · **Hardware**

The heart and circulation

YI-02

C3



Smart Classroom Energy Saver

Power & Energy · **Hardware**

Relays and automation

YI-03

C4



Stepper Motor Drawing Plotter

Mechanical Design · **Hardware**

Stepper motors and mechanisms

YI-04

Phase 2

Weeks 3-4

Robots & Security

Scanning robots, a Bluetooth car, a gesture arm and an RFID lock

C5



Obstacle Avoiding Robot with Scanning Sonar

Robotics & Actuation · **Hardware**

Autonomous navigation

YI-05

C6



Bluetooth Robot Car with Phone App

IoT & Communication · **Hardware + Software**

Serial communication and app design

YI-06

C7



Gesture-Controlled Robot Arm

Robotics & Actuation · **Hardware**

Robot arms and degrees of freedom

YI-07

C8



RFID Smart Door Lock with Access Log

Security & Safety · **Hardware**

Arrays, comparisons and logs

YI-08

Phase 3

Weeks 5-6

Power & Wireless

Battery packs, radio doorbells, a Wi-Fi plant system and its phone app

C9



Rechargeable Power Pack for Robots

Power & Energy · **Hardware**

Charging, protection and voltage conversion

YI-09

C10



Wireless Doorbell with Visitor Counter

IoT & Communication · **Hardware**

Radio communication and addressing

YI-10

C11



Wi-Fi Automatic Plant Watering System

Sensors & Environment · **Hardware**

Plants, soil and water use

YI-11

C12



Plant Monitor Mobile App

App, Web & Game Development · **Software**

Apps, web requests and events

YI-12

Phase 4

Weeks 7-8

Software, AI & Design

Python games, a portfolio website, a chatbot and a 3D-printed enclosure

C13



Python Maths Quiz Game with Scoreboard

Programming & Logic · **Software**

Python variables, loops, conditions, functions

YI-13

C14



Personal Portfolio Website

App, Web & Game Development · **Software**

HTML, CSS and JavaScript

YI-14

C15



School Helpdesk Chatbot

AI & Data · **Software**

Rule-based vs learning systems

YI-15

C16



3D-Printed Plant Monitor Enclosure

Mechanical Design · **Software**

Design for manufacture and tolerances

YI-16



After class 16 • Optional capstone showcase (outside the 16-class batch)

Suggested projects: Earthquake early-warning alarm; Wi-Fi smart mailbox notifier; Greenhouse ventilation controller; Scheduled pet feeder with phone app. Teams choose a problem, plan, build, get feedback, improve and present at a showcase.

CLASS
1

555 Timer Alarm Siren Circuit

Electronics & Circuits • Hardware • Week 1

YI-01

Introduction

Before microcontrollers, engineers built timers and sirens using a small chip called the 555 timer, still found in millions of devices today. Students build a flashing LED and a two-tone siren with a 555, capacitors and resistors, with no Arduino and no code. They test the circuit in the Falstad simulator, then build it on a breadboard and measure voltages with a multimeter. Turning a potentiometer changes the speed, showing how resistance and capacitance control time.

Activities

GUIDED

- Simulate the 555 astable circuit in Falstad
- Build it on a breadboard and measure with a multimeter

DISCOVERY

- Change the capacitor or potentiometer to make a faster or slower siren

Outcomes

Science: Capacitors, resistance and charging time**Engineering:** Integrated circuits without code**Maths:** Frequency and period**Software****Falstad Circuit Simulator**

Also: Tinkercad Circuits (has a 555)

Key hardware (per group)

- 555 timer IC x1
- Capacitor 10 uF x1
- Capacitor 10 nF x1
- Capacitor 100 uF x1
- Resistor 1k ohm x2
- Resistor 10k ohm x1
- Resistor 220 ohm x1
- + 4 more (see workbook)

Standards

NGSS MS-ETS1 • CSTA Computing Systems

SDG badgesCLASS
2

Heart Rate Monitoring System

Sensors & Environment • Hardware • Week 1

YI-02

Introduction

Doctors, athletes and smartwatches all track heart rate to spot health problems early. Students build a heart rate monitor with an optical pulse sensor: the Arduino detects each beat, flashes an LED in time and calculates beats per minute. A live pulse graph and BPM value appear on an OLED screen, and a buzzer warns if the rate goes too high or too low. They compare resting and active heart rates and discuss privacy of health data.

Activities

GUIDED

- Read the pulse signal and detect each beat
- Calculate BPM and draw a live graph on the OLED

DISCOVERY

- Add high and low alerts, then test after 1 minute of exercise

Outcomes

Science: The heart and circulation**Computing:** Signal thresholds, timing and averages**Life Skills:** Health data privacy**Software****Arduino IDE 2**

Also: Wokwi (OLED practice)

Key hardware (per group)

- Arduino Uno + USB cable x1
- Pulse sensor x1
- OLED 0.96 in I2C x1
- Buzzer x1
- LED 5 mm (assorted) x1
- Resistor 220 ohm x1

Standards

NGSS MS-ETS1 • CSTA Data & Analysis

SDG badges

CLASS
3

Smart Classroom Energy Saver

Power & Energy + Sensors & Environment • Hardware • Week 2

YI-03

Introduction

Schools waste energy when lights stay on in empty or bright rooms. Students design a controller that switches a lamp through a relay only when the room is occupied (PIR) and too dark (LDR). An OLED shows the room status and a running count of minutes the lamp was kept off. They present an estimate of the energy their design would save across the whole school in a year.

Activities

GUIDED

- Combine PIR and LDR readings with AND logic
- Switch a low-voltage lamp through the relay

DISCOVERY

- Add a 2-minute timeout so lights switch off after people leave

Outcomes

Engineering: Relays and automation**Computing:** Combined conditions (AND/OR) and timers**Maths:** Estimating energy savings**Software****Arduino IDE 2**

Also: Wokwi; Tinkercad Circuits

Key hardware (per group)

- Arduino Uno + USB cable x1
- PIR motion sensor x1
- LDR module x1
- Relay module (1-channel) x1
- Low-voltage LED lamp (5 V) x1
- OLED 0.96 in I2C x1

Standards

NGSS MS-ETS1 • CSTA Algorithms & Programming

SDG badgesCLASS
4

Stepper Motor Drawing Plotter

Mechanical Design + Programming & Logic • Hardware • Week 2

YI-04

Introduction

Printers, 3D printers and CNC machines move with stepper motors that turn in tiny exact steps. Students build a two-axis drawing machine from two stepper motors and a servo that lifts and lowers a pen. They program it to draw shapes by calculating how many steps each motor must take. The session connects mechanical design, coordinates and precise control. It may run as a double session.

Activities

GUIDED

- Drive each stepper a set number of steps
- Draw a square and a triangle using coordinates

DISCOVERY

- Program your initials or a simple logo

Outcomes

Engineering: Stepper motors and mechanisms**Maths:** Coordinates and ratios**Computing:** Functions for reusable moves**Software****Arduino IDE 2**

Also: Wokwi (stepper)

Key hardware (per group)

- Arduino Uno + USB cable x1
- Stepper 28BYJ-48 + ULN2003 x2
- Servo SG90 x1
- 4xAA battery pack x1

Standards

NGSS MS-ETS1 • ISTE Computational Thinker

SDG badges

CLASS
5**Obstacle Avoiding Robot with Scanning Sonar**

Robotics & Actuation • Hardware • Week 3

YI-05

Introduction

Self-driving cars scan their surroundings before choosing a direction. Students build a robot whose ultrasonic sensor sits on a servo, so it can look left, centre and right like a radar. When an obstacle is close, the robot stops, scans, compares distances and turns towards the clearest path. They write functions for each behaviour and test the robot in a maze of boxes.

**Software****Arduino IDE 2**

Also: ArduinoBlocks

Key hardware (per group)

- Arduino Uno + USB cable x1
- 2WD robot chassis kit x1
- L298N motor driver x1
- Ultrasonic sensor HC-SR04 x1
- Servo SG90 x1
- 4xAA battery pack x1

Standards

NGSS MS-ETS1 • CSTA Algorithms & Programming

SDG badges

Activities

GUIDED

- Mount the sensor on a servo and scan three directions
- Write functions: forward, stop, scan, turn

DISCOVERY

- Get through the obstacle course fastest without touching a wall

Outcomes

Engineering: Autonomous navigation**Computing:** Functions and comparisons**Technology:** How self-driving cars senseCLASS
6**Bluetooth Robot Car with Phone App**

IoT & Communication + App, Web & Game Development • Hardware + Software • Week 3

YI-06

Introduction

Remote-controlled vehicles, from toy cars to drones, send commands wirelessly. Students convert a robot chassis into a Bluetooth car and build their own Android controller in MIT App Inventor with direction buttons, headlights and a horn. The Arduino reads single-letter commands from the HC-05 module and drives the motors, LEDs and buzzer. They design the app layout for a real user and test it on a race course.

**Software****Arduino IDE 2 + MIT App Inventor**

Also: Serial Bluetooth Terminal

Key hardware (per group)

- Arduino Uno + USB cable x1
- 2WD robot chassis kit x1
- L298N motor driver x1
- Bluetooth module HC-05 x1
- LED 5 mm (assorted) x2
- Resistor 220 ohm x2
- Resistor 1k ohm x1
- + 3 more (see workbook)

Standards

CSTA Networks & the Internet • ISTE Innovative Designer

SDG badges

Activities

GUIDED

- Receive Bluetooth commands and drive the motors
- Build the controller app in MIT App Inventor

DISCOVERY

- Add a speed slider and a headlight button to your app

Outcomes

Computing: Serial communication and app design**Engineering:** Wireless control systems**Arts:** User interface design

CLASS
7**Gesture-Controlled Robot Arm**

Robotics & Actuation • Hardware • Week 4

YI-07

Introduction

Surgeons and bomb-disposal teams control robot arms from a distance, and some arms follow hand movements. Students build a three-servo arm from cardboard or ice cream sticks and control it by tilting a glove fitted with an accelerometer. They read tilt angles, map them to servo positions and smooth out shaky movements. They test by picking up a light object and discuss where gesture control is useful.

Activities

GUIDED

- Read tilt angles from the MPU6050
- Map tilt to three servo positions

DISCOVERY

- Add a button to open and close a gripper

Outcomes

Engineering: Robot arms and degrees of freedom**Maths:** Angles and mapping ranges**Computing:** Smoothing noisy sensor data**Software****Arduino IDE 2**

Also: Wokwi (MPU6050 + servo)

Key hardware (per group)

- Arduino Uno + USB cable x1
- MPU6050 IMU x1
- Servo SG90 x3
- Push button x1
- 4xAA battery pack x1

Standards

NGSS MS-ETS1 • ISTE Innovative Designer

SDG badges

Introduction

Offices, hotels and schools use RFID cards to decide who can enter. Students build a smart lock: the RFID reader checks each card's ID against a list, a servo unlocks the door for approved cards, and a buzzer and red LED reject unknown ones. An OLED shows who entered and keeps a short access log. They discuss privacy, lost cards and how to add or remove users safely.

Activities

GUIDED

- Read card IDs with the RC522
- Unlock the servo only for approved IDs, and log entries on the OLED

DISCOVERY

- Add a master card that can register new users

Outcomes

Computing: Arrays, comparisons and logs**Technology:** RFID in everyday life**Life Skills:** Privacy and access rights**Software****Arduino IDE 2**

Also: Wokwi (servo + OLED practice)

Key hardware (per group)

- Arduino Uno + USB cable x1
- RFID RC522 + cards x1
- Servo SG90 x1
- OLED 0.96 in I2C x1
- Buzzer x1
- LED 5 mm (assorted) x2
- Resistor 220 ohm x2

Standards

CSTA Impacts of Computing • ISTE Digital Citizen

SDG badgesCLASS
8**RFID Smart Door Lock with Access Log**

Security & Safety • Hardware • Week 4

YI-08

CLASS
9

Rechargeable Power Pack for Robots

Power & Energy + Electronics & Circuits • Hardware • Week 5

YI-09

Introduction

Every robot and phone needs a safe, rechargeable power source. Students build a power pack from an 18650 lithium cell, a TP4056 charger with protection, and an LM2596 converter that steps the voltage down for their robots. They adjust and verify the output with a multimeter and use an Arduino to show battery voltage on an OLED. Battery safety rules are taught and followed throughout, under teacher supervision.

Activities

GUIDED

- Wire the TP4056 charger and 18650 cell safely
- Set the LM2596 output with a multimeter and show voltage on the OLED

DISCOVERY

- Power your YI-05 robot from your own power pack

Outcomes

Engineering: Charging, protection and voltage conversion**Science:** Chemical energy storage**Life Skills:** Battery safety**Software**

Arduino IDE 2

Also: Falstad (voltage dividers)

Key hardware (per group)

- Arduino Uno + USB cable x1
- 18650 cell + holder x1
- TP4056 charger module x1
- LM2596 buck converter x1
- OLED 0.96 in I2C x1
- Resistor 10k ohm x2

Standards

NGSS MS-ETS1 • NGSS MS-PS3

SDG badgesCLASS
10

Wireless Doorbell with Visitor Counter

IoT & Communication • Hardware • Week 5

YI-10

Introduction

Wireless doorbells send a radio message from the front door to a chime inside. Students build one with two Arduinos and two NRF24L01 radio modules: pressing the outdoor button sends a message, and the indoor unit rings a buzzer and counts visitors on an OLED. They learn about addresses, sending and receiving, and test how far the signal reaches through walls. Pairs of groups work together on transmitter and receiver.

Activities

GUIDED

- Set up a transmitter and a receiver with NRF24L01
- Ring the chime and count visitors on the OLED

DISCOVERY

- Send the time of each visit and test the range through walls

Outcomes

Computing: Radio communication and addressing**Science:** Radio waves and range**Engineering:** Two-device systems**Software**

Arduino IDE 2

Key hardware (per group)

- Arduino Uno + USB cable x2
- NRF24L01 module x2
- Push button x1
- Resistor 10k ohm x1
- Buzzer x1
- OLED 0.96 in I2C x1
- Capacitor 10 uF x2

Standards

CSTA Networks & the Internet • NGSS MS-ETS1

SDG badges

CLASS
11

Wi-Fi Automatic Plant Watering System

Sensors & Environment + IoT & Communication • Hardware • Week 6

YI-11

Introduction

Water is scarce, and smart farms water crops only when the soil needs it. Students build an ESP32 system that reads soil moisture, temperature and humidity, switches a small pump through a relay when the soil is dry, and hosts a live status webpage on the school Wi-Fi. They set the watering threshold using evidence from their readings. This is their first internet-connected device.

Activities

GUIDED

- Read soil moisture and DHT22 with the ESP32
- Switch the pump through the relay and host a status webpage

DISCOVERY

- Stop watering if it is already humid, to save water

Outcomes

Science: Plants, soil and water use

Computing: Web servers on a microcontroller

Engineering: Pumps and relays



Software

Arduino IDE 2 (ESP32)

Also: Wokwi (ESP32)

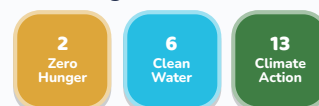
Key hardware (per group)

- ESP32 Dev Board + USB cable x1
- Soil moisture sensor x1
- DHT22 sensor x1
- Relay module (1-channel) x1
- Mini water pump + tube x1
- 4xAA battery pack x1

Standards

NGSS MS-ETS1 • CSTA Networks & the Internet

SDG badges


CLASS
12

Plant Monitor Mobile App

App, Web & Game Development • Software • Week 6

YI-12

Introduction

Most smart devices come with a phone app. Students design an Android app in MIT App Inventor that reads the YI-11 plant system over Wi-Fi and shows soil moisture, temperature and pump status with icons. A button lets the user water the plant from the phone. They plan screens on paper first, then build, test and improve the app with a classmate.

Activities

GUIDED

- Design app screens on paper
- Read data from the ESP32 webpage and display it

DISCOVERY

- Add a 'water now' button and a low-water warning

Outcomes

Computing: Apps, web requests and events

Arts: User interface design

Life Skills: Designing for a real user



Software

MIT App Inventor

Key hardware (per group)

- Laptop (1 per group)

Standards

CSTA Networks & the Internet • ISTE Creative Communicator

SDG badges



CLASS
13**Python Maths Quiz Game with Scoreboard**

Programming & Logic • Software • Week 7

YI-13

Introduction

Python is one of the world's most popular programming languages, used from games to AI. Students write a maths quiz game that asks random questions, checks answers, keeps score and saves a high-score table to a file. They use variables, loops, if-statements and functions. Classmates play each other's games and suggest improvements.

**Software****Python (Thonny)**

Also: Trinket

Key hardware (per group)

- Laptop (1 per group)

Standards

CSTA Algorithms & Programming • ISTE Computational Thinker

SDG badges**Activities****GUIDED**

- Generate random questions and check answers
- Keep score with variables and save high scores to a file

DISCOVERY

- Add difficulty levels and a timer

Outcomes**Computing:** Python variables, loops, conditions, functions**Maths:** Arithmetic practice**Life Skills:** Giving and using feedbackCLASS
14**Personal Portfolio Website**

App, Web & Game Development • Software • Week 7

YI-14

Introduction

Engineers show their work to universities and employers through online portfolios. Students build a personal website with HTML, CSS and a little JavaScript, including a page for each project they have built this term with photos and short explanations. They learn page structure, styling, layout and a simple interactive feature. The site becomes their digital record of the programme.

**Software****Online HTML editor (e.g. CodePen)**

Also: VS Code

Key hardware (per group)

- Laptop (1 per group)

Standards

ISTE Creative Communicator • CSTA Algorithms & Programming

SDG badges**Activities****GUIDED**

- Build pages with HTML and style them with CSS
- Add a JavaScript photo gallery or dark-mode button

DISCOVERY

- Write a clear 3-sentence explanation of your best project

Outcomes**Computing:** HTML, CSS and JavaScript**Arts:** Layout and visual design**Life Skills:** Presenting your work

CLASS
15**School Helpdesk Chatbot**

AI & Data • Software • Week 8

YI-15

Introduction

Chatbots answer questions on websites and in apps every day. Students write a rule-based chatbot in Python that answers questions about the school (timings, clubs, events) by matching keywords. They then compare it with a modern AI chatbot and discuss what each does well, where each goes wrong, and why AI answers must be checked. This introduces how AI systems are designed and their limits.

**Software****Python (Thonny)**

Also: Teachable Machine

Key hardware (per group)

- Laptop (1 per group)

Standards

UNESCO AI Competency (Understand/Apply) • CSTA AI concepts

SDG badges**Activities****GUIDED**

- Build a keyword-matching chatbot in Python
- Test it with classmates and improve its answers

DISCOVERY

- Compare your bot with a modern AI chatbot: what is different?

Outcomes**AI:** Rule-based vs learning systems**Computing:** Strings, lists and dictionaries**Life Skills:** Using AI responsiblyCLASS
16**3D-Printed Plant Monitor Enclosure**

Mechanical Design • Software • Week 8

YI-16

Introduction

A real product needs a case that protects its electronics and fits in its environment. Students measure the parts of their YI-11 plant system with a ruler, then design a snug enclosure with holes for the sensor cables and USB port in Tinkercad or BlocksCAD. They slice it in Cura and estimate print time and material. Printed cases are test-fitted and improved.

**Software****Tinkercad 3D**

Also: BlocksCAD; UltiMaker Cura

Key hardware (per group)

- Laptop (1 per group)

Standards

NGSS MS-ETS1 • ISTE Innovative Designer

SDG badges**Activities****GUIDED**

- Measure every part and sketch the enclosure
- Model it with openings and export it for Cura

DISCOVERY

- Improve the design after a test fit

Outcomes**Engineering:** Design for manufacture and tolerances**Maths:** Measurement and volume**Arts:** Product appearance



16-class roadmap

8 weeks, 2 classes a week: four phases of 4 classes. The capstone showcase runs after the batch.

Phase 1

Weeks 1-2

Power, Autonomy & Health

Off-grid solar IoT, a PID maze robot, a record-and-replay arm and a fall detector

C1



Solar-Powered Off-Grid Weather Node

Power & Energy • **Hardware**
Energy budgets and low-power design

TL-01

C2



Autonomous Maze Solving Robot

Robotics & Actuation • **Hardware**
Feedback control (PID)

TL-02

C3



Pick-and-Place Robotic Arm with Record & Replay

Mechanical Design • **Hardware**
Mechanisms, torque and degrees of freedom

TL-03

C4



Fall Detection Wearable for the Elderly

Sensors & Environment • **Hardware**
Acceleration and human movement

TL-04

Phase 2

Weeks 3-4

Connected Systems

ESP-NOW networks, a cloud smart home, smart parking and a two-factor locker

C5



ESP-NOW Wireless Sensor Network

IoT & Communication • **Hardware**
Wireless protocols and packets

TL-05

C6



Smart Home Cloud Control Hub

IoT & Communication • **Hardware**
Cloud IoT, MQTT and dashboards

TL-06

C7



Smart Parking Occupancy Dashboard

Sensors & Environment • **Hardware**
Web dashboards and real-time data

TL-07

C8



Two-Factor Smart Locker with Intrusion Alerts

Security & Safety • **Hardware**
Authentication and security logic

TL-08

Phase 3

Weeks 5-6

Data, AI & Web

Data pipelines, machine learning, AI ethics and REST APIs

C9



Sensor Data Logging & Analysis Pipeline

AI & Data • **Hardware + Software**
Data cleaning, visualisation and analysis

TL-09

C10



Machine Learning Rain Predictor

AI & Data • **Hardware + Software**
Supervised learning, train/test split, accuracy

TL-10

C11



AI Fake News Detector

AI & Data • **Software**
Text classification and bias

TL-11

C12



Web-Controlled Devices via REST API

App, Web & Game Development • **Hardware + Software**
REST APIs, HTTP and JSON

TL-12

Phase 4

Weeks 7-8

Security, Robots & Product Design

IoT security, a fire-fighting robot, parametric CAD and a motion-controlled game

C13



Securing an IoT Device

Security & Safety • **Software**
Cryptography, authentication and vulnerabilities

TL-13

C14



Autonomous Fire-Fighting Robot

Robotics & Actuation • **Hardware**
Search behaviour and actuators

TL-14

C15



Parametric Robot Gripper Design

Mechanical Design • **Software**
Parametric CAD and tolerances

TL-15

C16



Motion-Controlled Game

App, Web & Game Development • **Hardware + Software**
Serial communication and game loops

TL-16



After class 16 • Optional capstone showcase (outside the 16-class batch)

Suggested projects: Autonomous delivery robot; Full-stack IoT product with dashboard; Predictive plant health analytics; Assistive device for a person with a disability. Teams choose a problem, plan, build, get feedback, improve and present at a showcase.

CLASS
1

Solar-Powered Off-Grid Weather Node

Power & Energy + Electronics & Circuits • Hardware • Week 1

TL-01

Introduction

Remote weather stations in deserts and mountains run for months on solar power alone. Students design an off-grid node: a solar panel charges an 18650 cell through a TP4056, and an ESP32 wakes from deep sleep, reads a DHT22, sends the data over Wi-Fi and sleeps again. They measure current and battery voltage to calculate how long the node can run. The focus is energy budgeting, the key skill in low-power product design.



Software

Arduino IDE 2 (ESP32)

Also: Arduino Cloud

Key hardware (per group)

- ESP32 Dev Board + USB cable x1
- Small solar panel x1
- TP4056 charger module x1
- 18650 cell + holder x1
- DHT22 sensor x1
- Resistor 10k ohm x2

Standards

NGSS HS-ETS1 • NGSS HS-PS3

SDG badges



Activities

GUIDED

- Wire solar panel, charger, cell and ESP32
- Program deep sleep and a wake-read-send cycle

DISCOVERY

- Calculate battery life and improve it by changing the wake interval

Outcomes

Engineering: Energy budgets and low-power design**Science:** Solar energy and battery chemistry**Maths:** Power, current and time calculationsCLASS
2

Autonomous Maze Solving Robot

Robotics & Actuation + Programming & Logic • Hardware • Week 1

TL-02

Introduction

Rescue robots must find their way through unknown buildings without a map. Students build a robot with three ultrasonic sensors that follows walls using PID control, keeping a steady distance while turning at junctions. They tune the proportional, integral and derivative values and record how each change affects the robot's path. The robot then solves a cardboard maze using the left-hand rule. This may run as a double session.



Software

Arduino IDE 2

Key hardware (per group)

- Arduino Uno + USB cable x1
- 2WD robot chassis kit x1
- L298N motor driver x1
- Ultrasonic sensor HC-SR04 x3
- 4xAA battery pack x1

Standards

NGSS HS-ETS1 • CSTA Algorithms & Programming

SDG badges



Activities

GUIDED

- Read three ultrasonic sensors
- Implement PID wall-following and tune Kp, Ki, Kd

DISCOVERY

- Solve the maze and compare times after tuning

Outcomes

Engineering: Feedback control (PID)**Computing:** Algorithms: wall-following and maze solving**Maths:** Error, rates and tuning

CLASS
3**Pick-and-Place Robotic Arm with Record & Replay**

Mechanical Design • Hardware • Week 2

TL-03

Introduction

Factory robots are taught a task once by an operator and then repeat it thousands of times. Students build a four-servo arm and a matching controller of four potentiometers. They record a sequence of positions with a button, store it in memory (EEPROM), and play it back automatically to move objects from one place to another. They also calculate the arm's reach using basic trigonometry.

**Software****Arduino IDE 2**

Also: Wokwi (servos + pots)

Key hardware (per group)

- Arduino Uno + USB cable x1
- Servo MG995 x2
- Servo SG90 x2
- Potentiometer 10k x4
- Push button x2
- Resistor 10k ohm x2
- LM2596 buck converter x1
- + 1 more (see workbook)

Standards

NGSS HS-ETS1 • ISTE Innovative Designer

SDG badges

Activities

GUIDED

- Map four potentiometers to four servos
- Record positions and replay the sequence from EEPROM

DISCOVERY

- Program a pick-and-place task and measure its accuracy

Outcomes

Engineering: Mechanisms, torque and degrees of freedom**Computing:** Arrays and non-volatile memory**Maths:** Trigonometry for arm reachCLASS
4**Fall Detection Wearable for the Elderly**

Sensors & Environment + IoT & Communication • Hardware • Week 2

TL-04

Introduction

Falls are a leading cause of injury for older people, especially those living alone. Students build a wearable with an ESP32 and an accelerometer that recognises the sudden spike and stillness of a fall. It sounds a buzzer, gives the wearer 15 seconds to cancel with a button, and then sends a Wi-Fi alert to a caregiver's phone. They test with safe simulated falls and measure false alarms.

**Software****Arduino IDE 2 (ESP32)**

Also: Arduino Cloud (alerts)

Key hardware (per group)

- ESP32 Dev Board + USB cable x1
- MPU6050 IMU x1
- Buzzer x1
- Push button x1
- Resistor 10k ohm x1
- 18650 cell + holder x1

Standards

NGSS HS-ETS1 • CSTA Data & Analysis

SDG badges

Activities

GUIDED

- Record accelerometer data for walking, sitting and a simulated fall
- Detect a fall and send an alert with a cancel window

DISCOVERY

- Reduce false alarms and report your accuracy

Outcomes

Science: Acceleration and human movement**Computing:** Pattern detection and state machines**Life Skills:** Designing for vulnerable people

CLASS
5**ESP-NOW Wireless Sensor Network**

IoT & Communication • Hardware • Week 3

TL-05

Introduction

Smart buildings use many small wireless sensors that report to one central hub. Students build a network of three ESP32 nodes: one measures temperature and humidity, one light and one sound. Each sends its readings using the ESP-NOW protocol to a base station that shows a live summary on an OLED. They study range, packet loss and how networks handle a node going offline. This may run as a double session.

Activities**GUIDED**

- Program three sensor nodes and a base station with ESP-NOW
- Display all readings on the base OLED

DISCOVERY

- Measure packet loss at different distances

Outcomes**Computing:** Wireless protocols and packets**Engineering:** Distributed systems**Maths:** Reliability percentages**Software**

Arduino IDE 2 (ESP32)

Key hardware (per group)

- ESP32 Dev Board + USB cable x4
- DHT22 sensor x1
- LDR module x1
- Sound sensor KY-037 x1
- OLED 0.96 in I2C x1

Standards

CSTA Networks & the Internet • NGSS HS-ETS1

SDG badges**Introduction**

Smart homes let people control lights and appliances from anywhere and automate them with rules. Students build an ESP32 hub with a four-channel relay, a PIR sensor and a DHT22, connected to a cloud dashboard (Arduino Cloud or MQTT). From a phone they switch devices, view live sensor data and create rules such as 'turn on the fan above 30°C'. They discuss reliability and what happens when the internet goes down.

Activities**GUIDED**

- Connect the ESP32 to a cloud dashboard
- Control four relays and display PIR and DHT22 data

DISCOVERY

- Create two automation rules and a manual override

Outcomes**Computing:** Cloud IoT, MQTT and dashboards**Engineering:** Home automation systems**Life Skills:** Reliability and safety of smart devices**Software**

Arduino Cloud

Also: Node-RED / MQTT

Key hardware (per group)

- ESP32 Dev Board + USB cable x1
- Relay module (4-channel) x1
- PIR motion sensor x1
- DHT22 sensor x1
- Low-voltage LED lamp (5 V) x2
- 5 V DC fan x1

Standards

CSTA Networks & the Internet • ISTE Innovative Designer

SDG badges

CLASS
7

Smart Parking Occupancy Dashboard

Sensors & Environment • Hardware • Week 4

TL-07

Introduction

Drivers in busy cities waste time and fuel searching for parking. Students build a three-bay model car park with an ultrasonic sensor per bay and red/green LEDs. An ESP32 publishes the occupancy to a live web dashboard showing free spaces and how long each bay has been used. They analyse usage data and propose how a real school car park could use the system.



Activities

GUIDED

- Detect occupancy in three bays and light red/green LEDs
- Publish live occupancy to a web dashboard

DISCOVERY

- Add bay usage times and a 'car park full' alert

Outcomes

Computing: Web dashboards and real-time data

Engineering: Smart city systems

Maths: Usage statistics

Software

Arduino IDE 2 (ESP32)

Also: Arduino Cloud

Key hardware (per group)

- ESP32 Dev Board + USB cable x1
- Ultrasonic sensor HC-SR04 x3
- LED 5 mm (assorted) x6
- Resistor 220 ohm x6
- Resistor 1k ohm x3
- Resistor 10k ohm x3

Standards

CSTA Data & Analysis • NGSS HS-ETS1

SDG badges

Introduction

Banks and online accounts use two-factor authentication: something you have plus something you know. Students build a locker that opens only with the correct RFID card AND the right PIN on a keypad, using a servo lock. A PIR sensor detects tampering and the ESP32 sends an alert and keeps a log of every attempt. They analyse the design for weaknesses and improve it.



Activities

GUIDED

- Verify RFID card and keypad PIN together
- Log attempts and send tamper alerts over Wi-Fi

DISCOVERY

- Threat-model your locker: list three attacks and fix one

Outcomes

Computing: Authentication and security logic

Technology: Physical and cyber security

Life Skills: Ethical responsibility for security

Software

Arduino IDE 2 (ESP32)

Key hardware (per group)

- ESP32 Dev Board + USB cable x1
- RFID RC522 + cards x1
- 4x4 matrix keypad x1
- Servo SG90 x1
- PIR motion sensor x1
- Buzzer x1
- OLED 0.96 in I2C x1

Standards

CSTA Impacts of Computing • ISTE Digital Citizen

SDG badgesCLASS
8

Two-Factor Smart Locker with Intrusion Alerts

Security & Safety • Hardware • Week 4

TL-08

CLASS
9**Sensor Data Logging & Analysis Pipeline**

AI & Data • Hardware + Software • Week 5

TL-09

Introduction

Scientists and engineers make decisions from data, not guesses. Students log a week of classroom data (temperature, humidity, light, sound) from an ESP32 to a CSV file, then clean and analyse it in Python with pandas and matplotlib. They look for patterns such as when the room is noisiest or warmest and present evidence-based recommendations. This builds the data skills needed for machine learning.

Activities**GUIDED**

- Log sensor readings to CSV from the ESP32
- Clean, graph and summarise the data in Python

DISCOVERY

- Present one evidence-based recommendation for the classroom

Outcomes

AI: Data cleaning, visualisation and analysis

Computing: Python pandas and matplotlib

Science: Environmental patterns

**Software**

Python (pandas, matplotlib)

Also: Google Sheets + Looker Studio

Key hardware (per group)

- ESP32 Dev Board + USB cable x1
- DHT22 sensor x1
- LDR module x1
- Sound sensor KY-037 x1

Standards

UNESCO AI Competency (Apply) • CSTA Data & Analysis

SDG badges**Introduction**

Weather forecasters use models trained on past data to predict rain. Students collect temperature, humidity, light and rain-sensor data, label it, and train a simple classifier in scikit-learn to predict whether it will rain. They split data into training and test sets, measure accuracy and discuss why a model can be confident and still wrong. The best model is loaded back to give a live prediction.

Activities**GUIDED**

- Collect and label a dataset from the sensors
- Train and test a classifier, and measure accuracy

DISCOVERY

- Improve accuracy with more features or data and explain the change

Outcomes

AI: Supervised learning, train/test split, accuracy

Science: Weather and humidity

Maths: Probability and evaluation

**Software**

Python (scikit-learn)

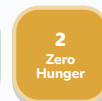
Also: Google Colab

Key hardware (per group)

- ESP32 Dev Board + USB cable x1
- DHT22 sensor x1
- LDR module x1
- Rain sensor x1

Standards

UNESCO AI Competency (Apply/Create) • CSTA AI concepts

SDG badges

CLASS
11**AI Fake News Detector**

AI & Data • Software • Week 6

TL-11

Introduction

False news spreads faster than true news online, and platforms use AI to flag it. Students train a text classifier in Python on a dataset of real and fake headlines, measure its accuracy and inspect the words it relies on. They then investigate bias: which topics or writing styles does it misjudge? They finish with a structured debate on whether AI should decide what news people see.

**Software****Python (scikit-learn)**

Also: Google Colab; Teachable Machine

Key hardware (per group)

- Laptop (1 per group)

Standards

UNESCO AI Competency (Ethics/Create) • ISTE Knowledge Constructor

SDG badges

Activities

GUIDED

- Prepare a labelled headline dataset and train a text model
- Evaluate accuracy and inspect influential words

DISCOVERY

- Find three cases where the model is unfair or wrong and explain why

Outcomes

AI: Text classification and bias**Life Skills:** Media literacy and ethics**Computing:** Model evaluationCLASS
12**Web-Controlled Devices via REST API**

App, Web & Game Development + IoT & Communication • Hardware + Software • Week 6

TL-12

Introduction

Behind every smart app is an API that lets programs talk to devices. Students write a REST API on an ESP32 that reports sensor data and switches a relay and LEDs, then build a web page and a Flask server that call it. They test endpoints, handle errors and document the API so another team can use it. This is full-stack development from hardware to browser.

**Software****Arduino IDE 2 + Flask**

Also: Postman

Key hardware (per group)

- ESP32 Dev Board + USB cable x1
- Relay module (1-channel) x1
- Low-voltage LED lamp (5 V) x1
- LED 5 mm (assorted) x2
- Resistor 220 ohm x2
- DHT22 sensor x1

Standards

CSTA Networks & the Internet • ISTE Creative Communicator

SDG badges

Activities

GUIDED

- Create GET and POST endpoints on the ESP32
- Build a web page that controls the device through the API

DISCOVERY

- Document your API and let another team control your device

Outcomes

Computing: REST APIs, HTTP and JSON**Engineering:** Full-stack systems**Life Skills:** Writing documentation for others

CLASS
13**Securing an IoT Device**

Security & Safety • Software • Week 7

TL-13

Introduction

Millions of smart devices are hacked every year because of weak passwords and unencrypted data. Students learn core cybersecurity ideas through beginner picoCTF challenges (ciphers, hashing, web basics), then audit the TL-06 smart home hub they built. They fix default passwords, add authentication and remove exposed data, documenting each vulnerability and fix. Ethical hacking rules are agreed before starting.

Activities

GUIDED

- Solve beginner picoCTF challenges
- Audit and harden the TL-06 hub

DISCOVERY

- Write a short security report: risks found, fixes made

Outcomes

Computing: Cryptography, authentication and vulnerabilities

Life Skills: Ethics and the law in hacking

Technology: IoT security

**Software****picoCTF**

Also: Arduino Cloud (hub settings)

Key hardware (per group)

- Laptop (1 per group)

Standards

CSTA Impacts of Computing • ISTE Digital Citizen

SDG badges

Introduction

Fire-fighting robots enter burning buildings so people do not have to. Students build a robot with three flame sensors that locates a candle flame, drives towards it and stops at a safe distance. A servo aims a small fan, switched by a relay, to blow out the flame. They test in a controlled arena under teacher supervision and measure how quickly the robot responds.

Activities

GUIDED

- Locate the flame by comparing three flame sensors
- Drive to the flame and aim the fan with a servo

DISCOVERY

- Improve response time and add a 'mission complete' signal

Outcomes

Engineering: Search behaviour and actuators

Computing: Sensor comparison and state machines

Life Skills: Fire safety

**Software**

Arduino IDE 2

Key hardware (per group)

- Arduino Uno + USB cable x1
- 2WD robot chassis kit x1
- L298N motor driver x1
- Flame sensor x3
- Servo SG90 x1
- Relay module (1-channel) x1
- 5 V DC fan x1
- + 1 more (see workbook)

Standards

NGSS HS-ETS1 • ISTE Innovative Designer

SDG badgesCLASS
14**Autonomous Fire-Fighting Robot**

Robotics & Actuation + Security & Safety • Hardware • Week 7

TL-14

CLASS
15

Parametric Robot Gripper Design

Mechanical Design • Software • Week 8

TL-15

Introduction

Engineers use parametric CAD so one design can be resized instantly for different jobs. Students design a servo-driven gripper for their TL-03 arm in BlocksCAD, using variables for jaw width, thickness and gear size. They check that parts fit the SG90 servo, slice the model in Cura and estimate cost and print time. Printed grippers are tested on the arm and redesigned if needed.

Activities

GUIDED

- Model a parametric gripper with variables
- Check fit, slice in Cura and estimate cost

DISCOVERY

- Print, test on the TL-03 arm and improve one parameter

Outcomes

Engineering: Parametric CAD and tolerances**Maths:** Variables, geometry and gear ratios**Arts:** Industrial design**Software****BlocksCAD**

Also: Tinkercad 3D; UltiMaker Cura

Key hardware (per group)

- Laptop (1 per group)

Standards

NGSS HS-ETS1 • ISTE Innovative Designer

SDG badges

Introduction

Game consoles use controllers full of sensors. Students build a physical controller from an Arduino, a joystick and buttons, which sends its readings over serial to a game they write in Python with Pygame. They design gameplay, scoring and difficulty, then run a playtest with classmates and improve the game from feedback. It brings hardware, software and design together.

Activities

GUIDED

- Send joystick and button data over serial
- Build a Pygame game that responds to the controller

DISCOVERY

- Playtest with classmates and improve one mechanic

Outcomes

Computing: Serial communication and game loops**Arts:** Game design**Engineering:** Human-computer interaction**Software****Python (Pygame)**

Also: Arduino IDE 2

Key hardware (per group)

- Arduino Uno + USB cable x1
- Joystick module x1
- Push button x2
- Resistor 10k ohm x2

Standards

CSTA Algorithms & Programming • ISTE Creative Communicator

SDG badges

Capstone: project-based learning (optional, after the 16 classes)

Gold Standard PBL,
PBLWorks

After the 16 classes, two optional extra sessions turn skills into a team project. They sit outside the 16-class batch and teachers plan them separately; this page sets the shared structure so every pathway ends with a public product.



Challenging problem

A real problem or question



Voice & choice

Teams choose what to build



Critique & revision

Peer feedback, then improve



Public product

Showcase for parents and school

Session	Focus	Output
Extra A	Define the problem and user, plan the design, choose parts from past kits, start building	Project plan + first prototype
Extra B	Test, get feedback, improve, rehearse and present at the showcase	Working prototype + 2-minute presentation

Suggested projects by pathway

Little Explorers • Ages 5-7

- Light-up invention for my bedroom
- Smart plant pot helper
- Makey Makey musical instrument
- Talking robot face with lights and sounds

Junior Makers • Ages 8-11

- Automatic pet feeder with a timer
- Smart study lamp with posture reminder
- Obstacle avoiding delivery car
- Classroom noise-level meter

Young Innovators • Ages 12-14

- Earthquake early-warning alarm
- Wi-Fi smart mailbox notifier
- Greenhouse ventilation controller
- Scheduled pet feeder with phone app

Tech Leaders • Ages 15-17

- Autonomous delivery robot
- Full-stack IoT product with dashboard
- Predictive plant health analytics
- Assistive device for a person with a disability

Simple assessment rubric (1-4)

Criterion	What we look for
Problem & user	Clear problem and who it helps
Build & function	Prototype works; wiring and code are tidy
Process	Evidence of testing, feedback and improvement
Communication	Explains how it works and what they learned
Teamwork	Shared roles and respectful collaboration